Why Do Retirees Retain Wealth?

Quasi-experimental Evidence on Key Motives

Emre Oral

University of Mannheim

September 30th, 2026

Research Seminar in Applied Economics

Motivation

- Population aging makes the behavior of retirees central for public policy
- In standard life-cycle models, households drawdown assets in retirement
- Empirical studies find little dissaving in old age

  $\rightarrow$ the *retirement-savings puzzle* (Laitner et al. 2018, French et al. 2023) ▸ LCM
- Two main explanations:
  - Precautionary savings against health-related costs
  - Bequests

  $\Rightarrow$ hard to empirically disentangle, both fit wealth patterns well

This Paper

Research question: Why do households hold onto wealth in retirement?

I. **Document and benchmark the puzzle**
   how far is observed drawdown from what a life-cycle model predicts?

II. **Precautionary motive against long-term care (LTC) costs**
   identification using statutory kinks in LTC copayment schedule

III. **Bequest motive**
   matched difference-in-differences exploiting changes in the set of heirs

Preview of Results

I. Dutch retirees draw down less than one-fifth of net wealth by the mean age at death
   a life-cycle model with longevity risk predicts above 75%

II. Precaution against long-term care costs is real, but bounded
   €0.45 of extra yearly saving per €1 of copayment exposure $\rightarrow$ at most $\approx 1/5$ of net wealth

III. Bequest motives are directed at specific heirs
   losing an heir: −€11.6k (5.5% of mean wealth); a new grandchild: +€3.4k

$\Rightarrow$ the standard motives leave most old-age wealth unexplained

Setting: the Netherlands

1. High quality administrative data on the universe of Dutch residents
   - 1.6 million households over 20 years
   - linked wealth, income, family and inheritance registers

2. Quasi-experimental variation to test the standard explanations
   - kink in the long-term care copayment schedule
   - life events that change the set of heirs

3. An extensive welfare state insures most risks retirees face
   - universal low cost health insurance *(max €635/yr vs uncapped - U.S.)*
     + generous pension system *(75% RR vs 40% RR in the U.S.)*

$\rightarrow$ precautionary motives should be weak: a lab where the U.S. explanations are muted

long-term care is the exception – copayments run to €30,000/yr

Related literature

- Explaining low dissaving rates in old age using structural models
  - LTC / medical expenditure risk Palumbo (1999), De Nardi et al. (2010, 2024), Ameriks et al. (2011, 2020), Nakajima and Telyukova (2024) ...
  - Bequest motive Lockwood (2018), Kvaerner (2023), Fella et al. (2024), De Nardi et al. (2024) ...
- Estate planning and inter-vivo transfers Kopczuk (2007), Sturrock et al. (2022) ...

Contribution: novel causal evidence on key motives

- quasi-experimental design for care expenditure risk: kink in the LTC copayment schedule
- direct test of bequest motives using variation in the set of heirs – registers link the universe of retirees to their heirs

Outline

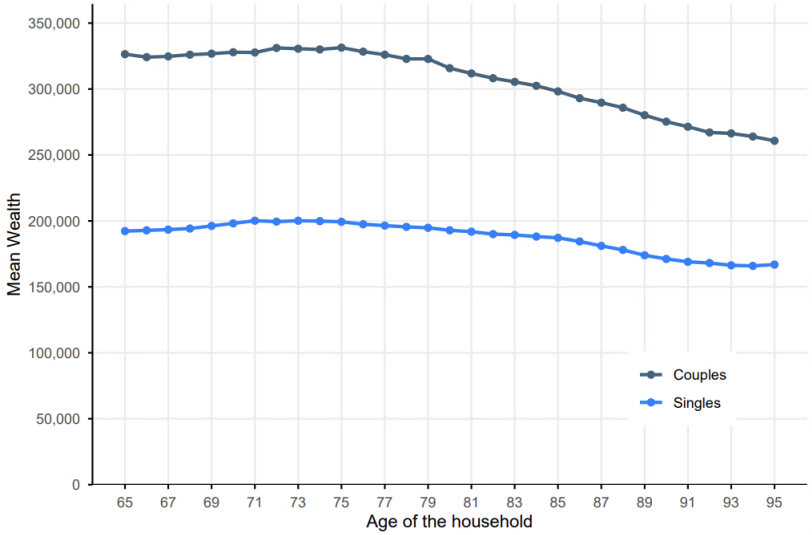
I. Wealth decumulation in old age

II. Precautionary motive against care expenditures

III. Bequest motive

Wealth – Age profile

Mean Wealth



- Cohort and Mortality
- Financial wealth
- Excluding Housing
- by birth cohort
- by age at death
- Median

How much drawdown should we expect?

	Observed in data	Life-cycle benchmark	
		Deterministic	Uncertain lifetime
Wealth drawn down by mean age at death	**16%**	**100%**	**77%**

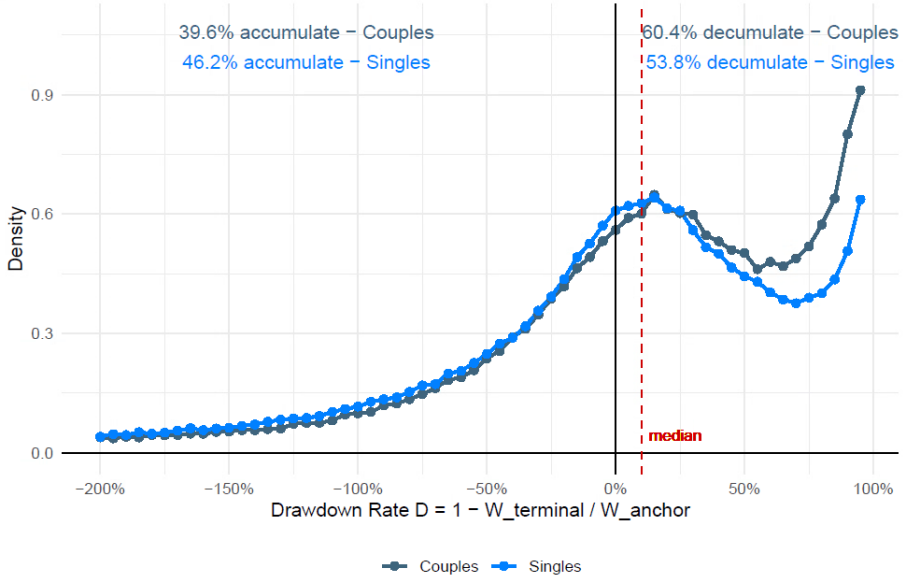
$$\ln W_{it} = \alpha_i + f(s_{it}) + \varepsilon_{it} \qquad D(s) = 1 - \exp\big(\hat{f}(s) - \hat{f}(0)\big)$$

s_{it}: years since retirement; α_i: household FE; $f(\cdot)$: cubic. $D(s)$: share of entry wealth drawn down.

Entry at 67 with one unit of wealth, mean age at death 86; CRRA utility, $\gamma = 4$, $r = 2\%$, no bequest motive. Pension income is a constant real flow of 5% of age-67 wealth per year, financed outside private wealth.

$\rightarrow$ Over two decades of retirement, households draw down barely a fifth of their wealth

Distribution of wealth drawdown



Summary: Decumulation Patterns

- Dutch retirees draw down 16% of their wealth by the mean age at death,
  against a 77% benchmark that prices in longevity risk
- Slow drawdown is broad-based, not driven by a few:
  the median household draws down 10%, and $\approx$ 45% of households *accumulate*
- Not an artifact of cohort or mortality composition

Factors that drive savings in retirement

1. Medical expenditure risk Palumbo (1999), De Nardi et al. (2010), Ameriks et al. (2020)
   out-of-pocket costs capped at €635/yr → lifetime exposure under 5% of net wealth

2. Care expenditure risk Ameriks et al. (2011), Nakajima and Telyukova (2024), De Nardi et al. (2024)
   the one uninsured tail: copayments run to €29,630/yr → **Section II**

3. Bequest motives Kopczuk (2007), Lockwood (2018), Kvaerner (2023)
   wealth is intended for heirs → **Section III**

▸ Details

⇒ the insurable risks are small here by construction – precaution has to come from care

Outline

I. Wealth decumulation in old age

II. Precautionary motive against care expenditures

III. Bequest motive

Identifying Savings Responses to Expected Care Costs

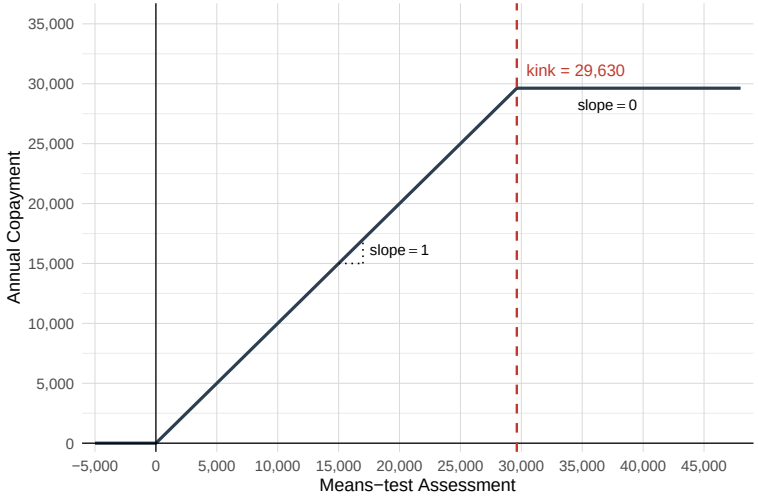
Challenge: finding quasi-random variation in expected care costs

Approach: exploit kink in long term care cost contribution schedule

- copayment is based on a means-tested component of income and wealth at t-2
- capped at a maximum: annually €29,630 in 2021

Identifying Savings Responses to Expected Care Costs

Approach: exploit kink in long term care cost contribution schedule



Empirical strategy: regression kink design (RKD)

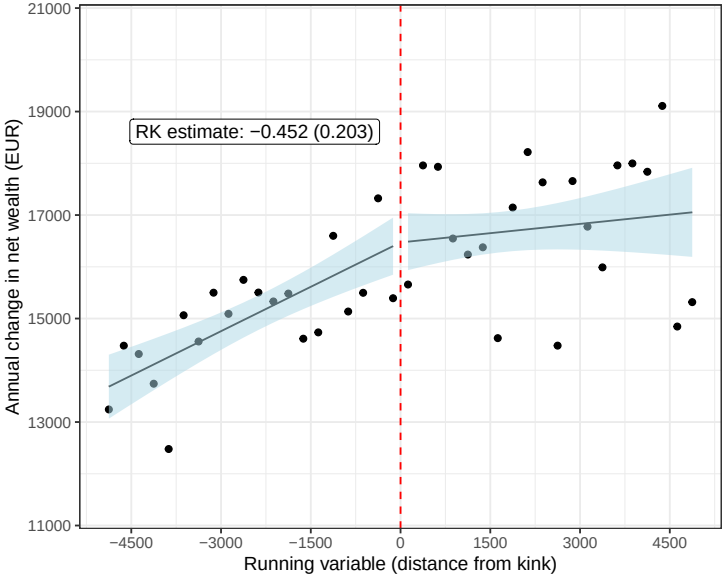
Estimate the change in slope of wealth accumulation around the kink:

$$\Delta W_{t,t+1,i} = \alpha + \gamma_o(r_i - k) + \boldsymbol{\gamma_1}\mathbb{1}[r_i > k](r_i - k) + \varepsilon_i \qquad \text{where } |r_i - k| < h$$

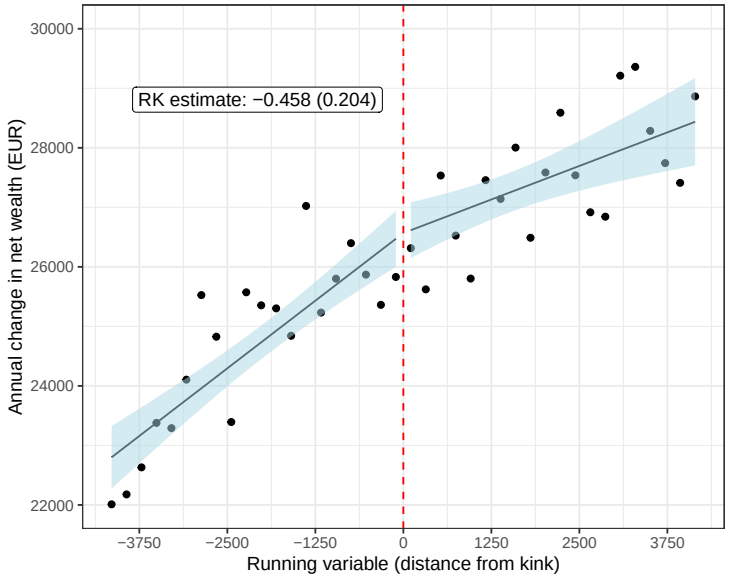
- $\Delta W_{t,t+1,i} = W_{t+1,i} - W_{t,i}$: change in net wealth,
- r_i: contribution base for household i,
- k: kink in the contribution schedule, h: bandwidth
- $\boldsymbol{\gamma_1}$: RKD estimate for marginal cost

Identification assumptions: (1) no manipulation of contribution base
(2) smooth wealth path absent policy

Main Results: Singles



Main Results: Couples



Identification Checks and Robustness

- Density of running variable: no discontinuity in means tested component around kink
  - ▸ Density around the kink
- RKD bandwidth choice
  - ▸ Bandwidth sensitivity
  - ▸ CCT optimal bandwidth estimates
- Placebo tests using artificial kinks
  - ▸ Kink $-$ 5000
  - ▸ Kink $+$ 5000
  - ▸ All placebos
- Covariate balancing
  - ▸ Age

Interpretation: marginal response

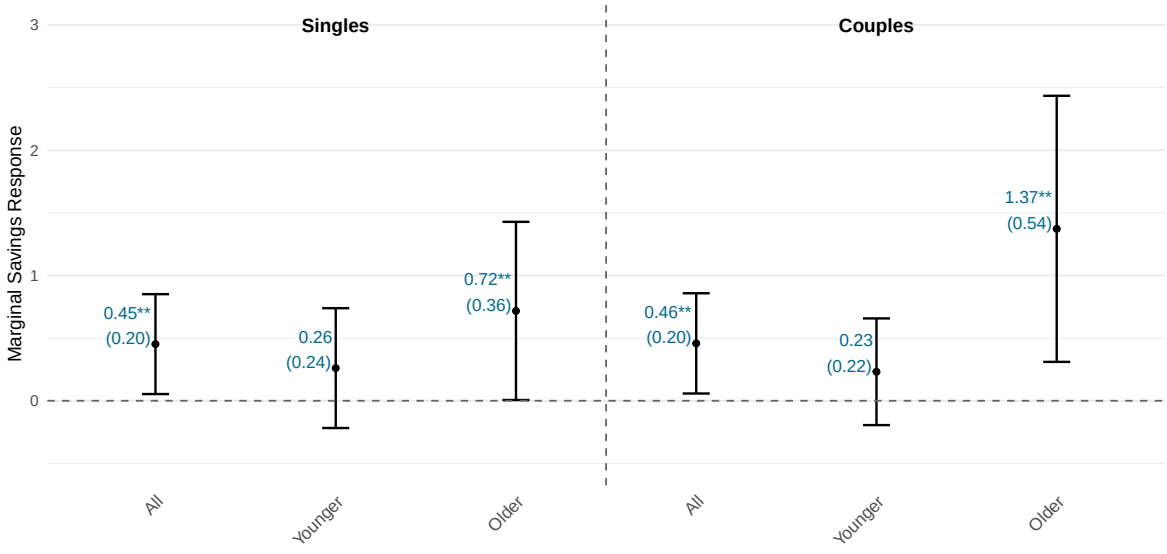
Dividing $\hat{\gamma}_1 = -0.45$ by change in slope of copayment gives the flow:

$$\hat{\beta}_{RKD} = \frac{\hat{\gamma}_1}{\Delta(\frac{dOOP}{dr})} = \frac{\hat{\gamma}_1}{0-1} = -\hat{\gamma}_1$$
$$= \frac{-0.45}{-1} = 0.45$$

$\hat{\beta}_{RKD}$: in response to €1 increase in out-of-pocket cost,
households increase yearly wealth by €0.45 around the kink

Heterogeneity by age

▸ RK plots by age



Interpretation: precautionary stock

Calculate the precautionary stock by accumulating the flow over the saving horizon:

$$W_{precautionary} = \hat{\beta}_{RKD} \times Cost \times \underbrace{\sum_t \frac{S(t)}{(1+r)^t}}_{\text{saving horizon}}$$

where $Cost$: max annual conditional-on-care OOP (€29,630);
$S(t)$: survival to year t of the saving horizon; r : real return

- Accumulate flow from average age in sample to average age of care entry
- $\hat{\beta}_{RKD}$ absorbs beliefs: household entry probability, expected care duration
- Main assumption: **linearity of marginal response** over the total schedule

Interpretation: precautionary stock

	Singles		Couples	
Version	Stock	% wealth	Stock	% wealth
Baseline	€90,800	28%	€92,000	26%
+ Survival to entry	€71,000	22%	€85,800	24%
+ Discounting ($r = 3\%$)	€64,400	20%	€77,300	21%

% wealth = stock / mean net wealth (Singles €319,645; Couples €359,590). Cost *Cap* =€29,630 (2021)

→ LTC precaution accounts for $\approx 1/5$ of wealth

▸ Figure

- Calculation likely yields an upper bound
  - Assumes all households pay the annual maximum
  - $\hat{\beta}_{RKD}$ is identified off households that are most exposed

Outline

I. Wealth decumulation in old age

II. Precautionary motive against care expenditures

III. Bequest motive

What a bequest motive implies

Retiree splits resources between own consumption and bequests to heirs:

$$U \;=\; u(c) \;+\; \sum_{k \,\in\, \text{children}} \theta_k \, v(b_k) \;+\; \sum_{j \,\in\, \text{others}} \theta_j \, v(b_j)$$

At the optimum, allocate until $u'(c) \;=\; \theta_m \, v'(b_m)$ for every heir m with $b_m > 0$

- **lose an heir** (child death) $\rightarrow$ $c \uparrow$, wealth $\downarrow$
- **gain an heir** (grandchild birth) $\rightarrow$ $c \downarrow$, wealth $\uparrow$

Only requires $\theta_k > 0$: the child is one weighted heir.
Does *not* require parents to weight children above other heirs.

Empirical Strategy: Matched Difference-in-Differences

▸ Event Study

Matching variables: wealth and income decile, household composition, marital status, age of eldest child, three years before the event

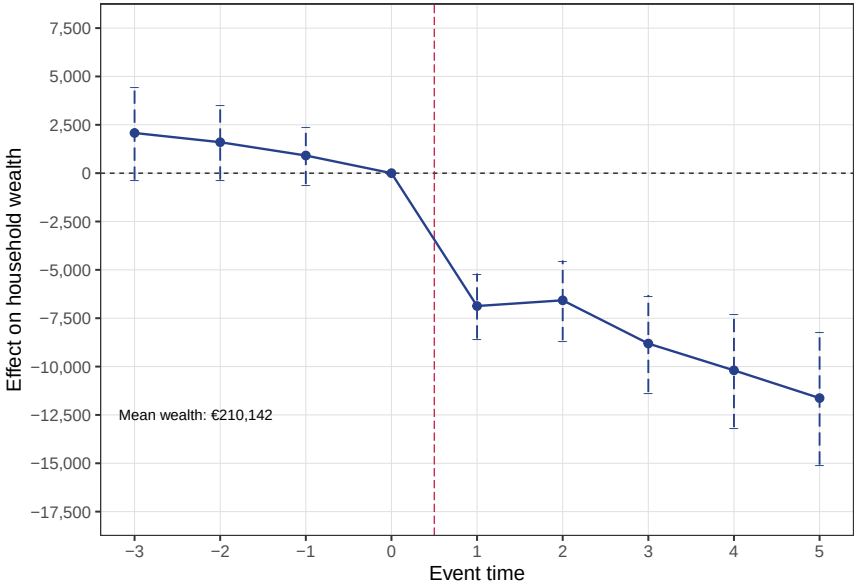
$$y_{it} = \alpha_i + \sum_{j \neq 0} \gamma_j D_{i,t-j} + \sum_{j \neq 0} \beta_j \cdot (\mathit{Treat}_i \cdot D_{i,t-j}) + \varepsilon_{it} \tag{1}$$

- y_{it} - wealth of household i at calendar time t
- Treat_i - household experiences a child life event (vs. matched control)
- $D_{i,t-j}$ - indicator for event time j, $j = -3, \ldots, +5$; $j = 0$ is the reference
- α_i and γ_j household and event-time fixed effects, ε_{it} idiosyncratic error
- β_j - coefficient of interest

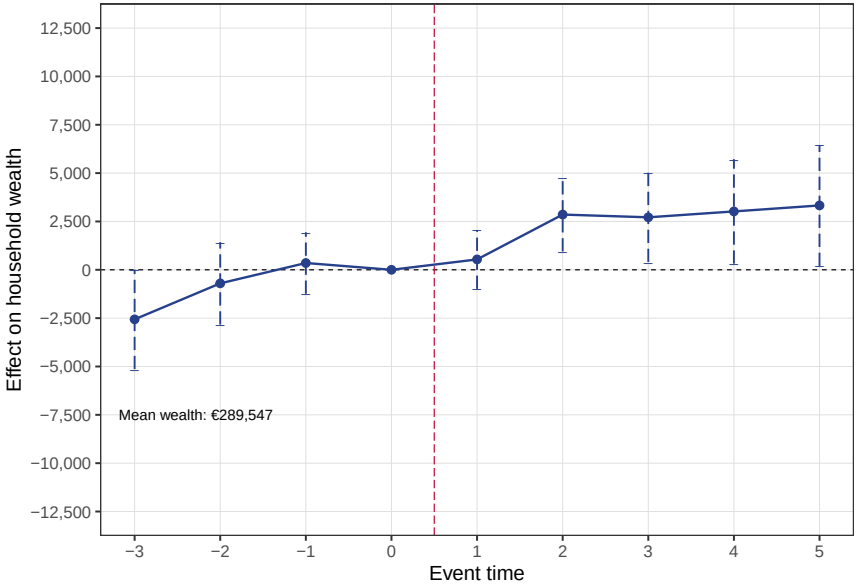
Restrictions:

- Child death event: require child to be without children
- Grandchild birth: require child and their spouse to be above 40 (anticipation of grandchild birth)

Losing an heir: death of a child



Gaining an heir: birth of a grandchild



Identification Checks and Robustness

- Wealth raw time series around changes in the beneficiary set ▸ Raw series
- Alternative specifications ▸ By specification
- Covariate balance ▸ Balance at the matching anchor
- Match rates ▸ Match rates

Interpretation: intended bequest share

	Death of a child		Birth of a grandchild		Total
	$\hat{\beta}_i$	Implied share of wealth (%)	$\hat{\beta}_i$	Implied share of wealth (%)	Implied share of wealth (%)
	(1)	(2)	(3)	(4)	(5)
Effect at $k = 5$	−11,630	14.0	3,328	3.5	17.5
	(1,756)	(2.1)	(1,597)	(1.7)	(2.7)
Mean wealth, $\hat{\mu}_i$	€210,142		€289,547		
Mean number of beneficiaries, N_i	2.53		3.05		

Opposite signed response points to an *intentional* bequest motive.

as opposed to incidental as by-product of precautionary savings or accidental from unexpected lifetimes

Conclusion

- Dutch retirees draw down barely a fifth of their wealth over two decades of retirement, against a 77%-100% benchmark.
- Precaution against care costs can explain up to $\approx 1/5$ of the wealth holdings.
- Evidence points to an *intentional* bequest motives.

  losing one heir: −€11.6k (-5.5% of mean wealth)
  gaining a grandchild: +€3.4k (+1.2% of mean wealth)

$\Rightarrow$ In the Netherlands, the two leading explanations explain up to 40$ of wealth held at old age,

but leave a significant share of old-age wealth unaccounted for.

The gap would be consistent with undirected bequest motives or behavioral explanations wealth in utility, habits, financial literacy

Thanks for listening!

emre.oral@uni-mannheim.de